Panama analysis guide

Version 0.2.5 | Matrix export and analysis tools

Complete lane review, alignment, baseline correction and shared sections first. Then open **06 Analysis**. Choose a method from the menu, adjust its settings and click **Run**. All analyses use the included samples and enabled sections.

1 Set up samples and groups

Click **Samples...** to edit sample names, biological groups, Pair IDs and inclusion in one table. Click **Apply**. Use group names such as Control and Treatment only when they match the experiment. K-means clusters do not overwrite these labels.

Choose **Measure**, **Normalize**, **Scale** and optional **log(1 + signal)**. Area is integrated corrected signal. Total signal normalization changes measurements into within-sample proportions; Gel standard requires a valid reference standard. Unit variance gives each varying section equal variance.

Mean signal (0-255) reports corrected linear intensity in 8-bit-equivalent units, keeping source precision. Individual 8-bit pixels are integers; averages can be decimals (100 and 101 average to 100.5). Background correction and interpolation also produce fractions. It multiplies normalized image means by 255; declared 0-255 numeric profiles are already on that scale. OD and profiles with unspecified units are unavailable. Existing Mean and Nirvana peak value (100 x mean in source units) remain available. Select numeric profile units during import or under Sample; this declares units without changing the source values.

2 Save the matrix as Excel

Click **Excel...**, choose a folder and filename, then save. Run PCA first if you want its exact input matrix and results included. Excel export also works without PCA.

Measurements contains the selected measurement before normalization or logging. **Analysis input** contains normalized/logged values. **PCA matrix** contains the exact centered/scaled values used for PCA, with constant sections removed. Rows are samples; columns after the identifying fields are sections.

The workbook also includes sample labels, exclusions, section bounds, settings and completed analyses. LDA fitted scores and cross-validated predictions are separate sheets. Blank numeric cells mean unavailable results. Excel is a static snapshot: edit the experiment and export again to update it.

3 PCA and heatmaps

Choose **PCA > Run PCA**. Use Scores, Loadings, Scree or Clusters to inspect the result. Group labels color the samples but do not influence the PCA fit.

Choose **Heatmap**. Set **Values** to Section z scores or Input values, and **Order** to Cluster rows + columns or Original order. Click **Run Heatmap**. Z scores show each section relative to its own mean; input values preserve the chosen normalization/log scale. Hover over a cell for its value. The standalone heatmap retains constant sections.

4 K means clustering

Choose **K-means**, set **Clusters K** and **Seed**, then run. K must be smaller than the sample count. The fit uses all varying sections with the chosen scaling; the plot is a two-axis PCA projection. Inspect membership, silhouette, inertia and the elbow table. Excel includes assignments and centroids. The same input, K and seed reproduce the fit.

Groups statistics and saving

5 Linear discriminant analysis

In **Samples...**, label every included sample. LDA needs at least **two groups** and **three independent samples per group**. Choose **LDA**, set the **Validation seed** and click **Run LDA**. Two groups produce one discriminant axis; additional groups can produce more.

The plot shows axes fitted using all labeled samples. Accuracy, balanced accuracy, the confusion table and predicted groups use **stratified cross-validation**, with up to five folds. Preprocessing is learned inside each training fold. The confusion table has actual groups in rows and predicted groups in columns. These are internal validation results, not proof of performance on new experiments. Repeated Pair IDs are rejected because this LDA uses independent-sample validation.

The **Validation seed** determines which samples are held out in each fold. The same seed repeats the split; a different seed can change accuracy and predictions, but does not change the plotted LD coordinates for the same data. Keep a fixed seed; do not pick one to maximize accuracy.

6 Descriptive statistics and group comparisons

Choose **Statistics**. **Descriptive** works with or without groups and reports n, mean, SD, SEM, median, minimum, maximum and a 95% confidence interval for each group mean. A single observation has no SD, SEM or mean confidence interval.

For comparisons, label all included samples and use at least two independent observations per selected group. Choose the test appropriate to your design:

Welch t test: compare the means of Group A and Group B without assuming equal variances. Select both groups. Mean difference and its 95% interval are A minus B.
Welch ANOVA: compare means across all labeled groups.
Mann-Whitney: rank-based comparison of two selected groups.
Kruskal-Wallis: rank-based comparison across all labeled groups.

Click **Run Statistics**. Each section has a test statistic, p value and **q (BH)**, adjusted across valid section tests in that run. Undefined tests stay blank with a reason. Omnibus tests do not identify which groups differ. Rank tests do not automatically establish a difference in medians.

Statistics use normalized/logged measurements **before centering or scaling**. The comparisons assume independent biological samples. Do not count duplicate technical lanes as independent replicates. For matched observations, use the paired tests on the next page. More than two repeated conditions need a separate repeated-measures model. Multiple exploratory runs need their own wider testing plan.

Paired comparisons and saving

7 Match paired samples

Open **Samples...**. Give both observations from the same subject or matched unit the same **Pair** ID, such as P01. Use different **Group** labels for the two conditions, such as Before and After. Each selected group must contain exactly one included observation per Pair ID. Click **Apply**.

Example: P01 / Before and P01 / After form one pair; P02 / Before and P02 / After form another. Pairing is based on the ID, never row order or lane number. Pair IDs must be unique across the experiment for each condition. Missing, duplicate or unmatched IDs stop the calculation; samples are never silently dropped.

Choose **Statistics**, then **Paired t test** or **Wilcoxon paired**. Select **Group A** and **Group B**, then run. At least two complete, independent pairs are required. Check the **Matched pairs** table before interpreting the results.

Paired t test tests the mean within-pair difference and reports its 95% confidence interval. It assumes normally distributed differences. **Wilcoxon paired** is the two-sided signed-rank test, with zero differences omitted and automatic p-value calculation. Its interpretation assumes a symmetric distribution of differences. Mean A minus B is shown descriptively; no mean confidence interval is claimed for Wilcoxon.

The same section-wise BH correction applies. Tests with no usable variation report an undefined result. Excel includes a **Paired matches** sheet, Pair IDs in **Samples**, and pair counts in **Group comparisons**. A paired test does not turn technical duplicates into independent biological replicates. More than two repeated conditions are not modeled here.

8 Save and reopen

Save experiment stores source images, edits, settings, group labels, Pair IDs and completed analyses in a .panama file. **Load experiment** restores them; **Save as...** creates a separate copy. Data edits clear stale results. Rerun the affected analyses before saving or exporting. **Export** produces the CSV/JSON bundle and includes Excel when analyses are available.

9 Export a graph

Open a graph, click **Export graph...**, choose **SVG**, **PNG** or **PDF**, then select **White** or **Dark** background. Click **Save...** and choose the folder and filename. White exports use dark labels and axes. Uncheck **Sample labels** to hide sample names in the export (also heatmap rows and dendrogram leaves). Points, axes, section labels and group legends remain. The on-screen graph is unchanged. SVG remains editable vector artwork; PNG and PDF use a lossless 300 dpi image. Export does not change measurements or the experiment.

10 Point shapes and colors

Use the **Sample labels** checkbox above the graph to show or hide sample names in the viewer. It applies to PCA Scores, LDA, K-means, both heatmaps and dendrogram leaves. Axes, section names, group legends and numerical results stay unchanged. Save experiment remembers the setting; Undo and Redo also work. Export graph starts with this viewer setting, and its own Sample labels checkbox can override it for that export.

Run **PCA (Scores)**, **LDA** or **K-means**, then click **Appearance...** beside **Export graph...**. Choose a shape or color in the **All groups** row to apply it to every group. Use an individual group row to override just that group. A small symbol previews each choice. Click **Apply**; **Cancel** keeps the previous appearance.

Eight shapes are available: circle, square, triangle, diamond, inverted triangle, plus, cross and star. The color picker accepts any color. **Reset** returns the current group set to circles and the automatic palette; click Apply to keep that reset. Choosing an All groups setting replaces individual overrides for that setting.

PCA and LDA share biological-group styles. K-means styles apply to cluster numbers separately, so they do not change biological-group colors. Appearance changes do not recalculate measurements, move points or change results. They support Undo and Redo, are stored by Save experiment, and appear in SVG, PNG and PDF exports, including legends and white-background exports.

Using the Mm01a example

The reviewed experiment has **18 samples and 14 sections**. With Area, no normalization, mean centering and no log transform, PCA gives **76.7% for PC1** and **13.3% for PC2**. You can also run its heatmap, K-means and descriptive statistics immediately. Biological group labels were not supplied, so assign the real groups before LDA or comparisons. Do not treat K-means clusters as independently defined biological groups.